

Spectral Normalization and Capacity Transport

Sharp Gram-Comparison Exponents and the Cost of Separate Factors

Prime Dynamics Theory Program

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Abstract

Suppose $mK^*K \preceq K'^*K' \preceq MK^*K$ and put $r = m/M$. We prove sharp cross-level factors: $q'_4 \geq r^{1/2}q_4$, normalized four-volume $\nu'_4 \geq r^{3/2}\nu_4$, and three-mode capacity $r\Lambda_{23} \leq \Lambda'_{23} \leq r^{-1}\Lambda_{23}$. Consequently, transporting volume and capacity as independent black boxes yields only $r^{5/2}q_4$, an extra loss r^2 relative to direct fourth-mode transport. All powers are sharp in separate diagonal families. A 4,096-instance audit has zero failures. This theorem fixes the normalization ledger but assumes, rather than proves, physical all-level Gram comparability.

1 Normalized quantities

Let $s_1 \geq s_2 \geq \dots$ be singular values and set

$$q_4 = \frac{s_4}{s_1}, \quad \Lambda_{23} = \frac{s_2 s_3}{s_1^2}, \quad \nu_4 = \frac{s_2 s_3 s_4}{s_1^3} = \Lambda_{23} q_4.$$

The last identity is the RH-110 factor ledger. Two-sided Gram comparison implies $\sqrt{m}, s_j(K) \leq s_j(K') \leq \sqrt{M}, s_j(K)$ by ordered eigenvalue monotonicity [Horn and Johnson, 1991, Bhatia, 2007].

Theorem 1 (Sharp normalized transport). *Under $mK^*K \preceq K'^*K' \preceq MK^*K$, with $r = m/M$,*

$$r^{1/2}q_4 \leq q'_4 \leq r^{-1/2}q_4, \\ r\Lambda_{23} \leq \Lambda'_{23} \leq r^{-1}\Lambda_{23}, \quad r^{3/2}\nu_4 \leq \nu'_4 \leq r^{-3/2}\nu_4.$$

Every displayed one-sided exponent is sharp.

Proof. Insert the singular-value bounds into each quotient after cancelling the common leading factor. Sharpness follows by diagonal scalings: use $\sqrt{M}$ on s_1 and $\sqrt{m}$ on s_4 for q_4 ; use $\sqrt{M}$ on s_1 and $\sqrt{m}$ on s_2, s_3, s_4 for ν_4 ; reverse the first three choices for the capacity upper. The opposite bounds are analogous. $\square$

Corollary 2 (Separate-factor penalty). *If one forgets the common operator and combines only $\nu'_4 \geq r^{3/2}\nu_4$ with $\Lambda'_{23} \leq r^{-1}\Lambda_{23}$, then*

$$q'_4 = \frac{\nu'_4}{\Lambda'_{23}} \geq r^{5/2}q_4.$$

This is weaker than the direct $r^{1/2}$ bound by exactly r^2 .

The corollary is not a contradiction: separate sharp bounds need not be simultaneously attained. It quantifies the price of discarding correlation.

2 Audit and consequence

We sample 4,096 source spectra, condition ratios spanning six decades, and independent singular scalings inside $[\sqrt{m}, \sqrt{M}]$. There are zero q_4 , capacity, or volume failures. Three diagonal records attain the corresponding sharp factors. RH-124 shows that a combined

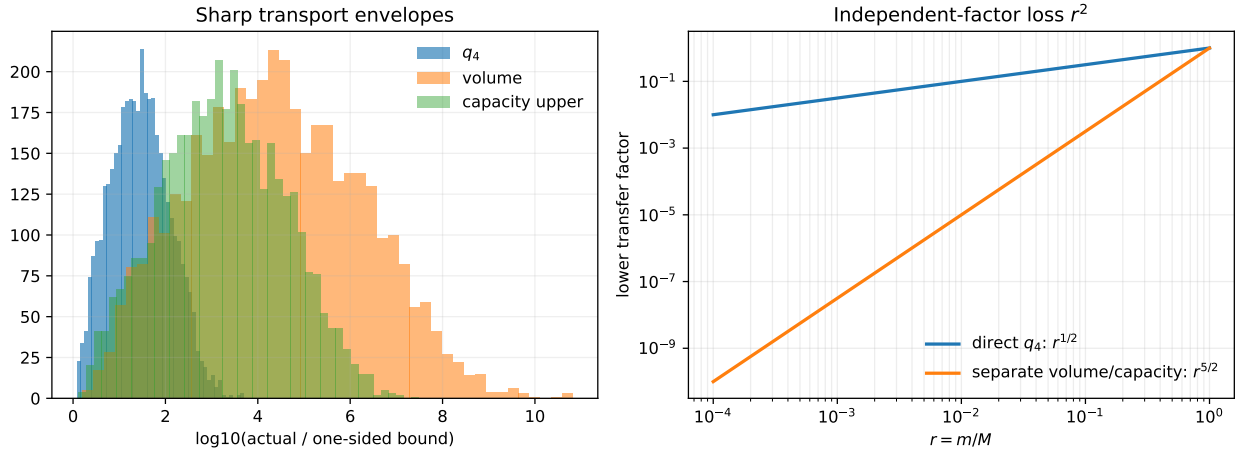


Figure 1: Transport-envelope efficiency and the r^2 loss caused by separate factorization.

directional theorem should preserve common cross-level information whenever possible. It does not supply physical m, M constants. No uniform Stage A, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis conclusion is claimed.

References

- Rajendra Bhatia. *Positive Definite Matrices*. Princeton University Press, 2007.
- Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.